

SUMMARY

Software Engineer with 6+ years of experience building web applications across Insurance, Finance, Banking, Manufacturing, and E-commerce. Deep expertise in React, Lit, Next.js, TypeScript, and Node.js — from design-system architecture and microfrontend platforms to AI-assisted component generation via multi-agent LLM orchestration. Currently leading the transformation of Caterpillar's enterprise design system (Blocks) from library-adoption to spec-driven AI generation. Proven impact: 60% bundle-size reduction, reusable component libraries serving 30+ teams, and cross-functional initiatives bridging design, engineering, and AI.

CORE COMPETENCIES

React TypeScript Next.js Node.js Lit / Web Components Design Systems LLM / GenAI Orchestration

MCP (Figma, Azure DevOps) Python OpenAI / Claude API Prompt Engineering Vector Databases Microfrontend

Angular Redux GraphQL Webpack / Vite Storybook Jest / Playwright AWS (Lambda, S3, Bedrock)

Azure DevOps Docker CI/CD SQL Git

PROFESSIONAL EXPERIENCE

Caterpillar Inc. — Software Engineer II

June 2025 – Present

Blocks — AI-Assisted, Spec-Driven Component Generation **IN PROGRESS**

Transforming Caterpillar's enterprise design system from library-adoption to AI-assisted generation, eliminating direct dependency for application teams.

- Architecting a multi-agent pipeline (5 producer agents) spanning Figma inspection → Azure DevOps sync → spec generation → React component output, with config-driven dispatch and per-component status tracking via JSON manifest.
- Designed a 10-layer quality framework with 77 automated checks plus Playwright runtime validation, enabling SME teams (Security, A11y, Performance) to independently own and version layer specs.
- Defined a framework-agnostic spec.yaml contract — a single spec generates parity components across React, Angular, and future stacks.
- Architected a Style Dictionary with multi-theme support automated from Figma design tokens, and designed HLDs for reusable components across the system.
- On track for ~70% reduction in time-to-component and 2.8x improvement in first-pass validator pass rate for pilot teams.

Stack: Claude/Harness AI agents, MCP (Figma, Azure DevOps), Lit Elements, React, Angular, Next.js, CSS Modules, Playwright, JSON Schema, YAML, Storybook, AWS, Azure, Nx, PNPM, Node.js

Blocks Frames — Enterprise Web Component Shell

- Developed helios-frames, an enterprise Lit Element library providing reusable Header, Footer, Profile, and Navigation components — reducing code redundancy by 30%.
- Integrated multi-auth system supporting B2C and Entra ID; built a monorepo with React, Angular, and Next.js playgrounds plus hybrid localization.

Stack: Lit Elements, React, Angular, Next.js, Web Components, Storybook, AWS, Azure, Nx, PNPM

EPAM Systems — Software Engineer II

July 2022 – May 2025

Everest Reinsurance — Insurance Underwriting Platform

Dashboards for Clients, Brokers, Underwriters, and Risk Managers to submit and approve contract forms.

- Led frontend architecture discovery; bootstrapped a production React TypeScript app with Webpack, Redux middleware, i18n, and CI/CD on Azure GitLab.
- Drove 60% bundle-size reduction and 40% faster initial load via code splitting, tree shaking, and lazy loading.
- Built reusable component library, custom hooks, and utility functions with full unit-test coverage.

Stack: React, TypeScript, Webpack, Storybook, AG-Grid, Material UI, Styled Components, SQL

LSEG Codebook — Jupyter Notebook Extensions for Data Science

Web application extending Jupyter Notebook for data science and ML professionals in investment banking.

- Built 10+ JupyterLab extensions and a library of 10+ React widgets; managed a Python package for frontend rendering and version releases.
- Collaborated across backend, cloud, and DevOps teams to troubleshoot and optimize deployments.

Stack: JupyterLab, React, Vue.js, TypeScript, Webpack, Docker, Jenkins, Jest, AWS

E-commerce Microfrontend Platform

- Implemented a microfrontend-based Next.js e-commerce application supporting payments, checkout, and CRED features following MACH principles.

Stack: Next.js, Frontastic, Commerce Tools, Adyen, GraphQL, Contentful

Discover Financials — Card Member Assistance (CMA)

Banking platform for account and card origination, management, and servicing for US card members.

- Developed reusable React components and UI templates used across multiple banking interfaces; involved in full SDLC.
- Optimized performance with React Hooks; implemented ADA accessibility compliance; managed CI/CD via Jenkins and GitLab.

Stack: React, Redux Toolkit, TypeScript, SCSS, Bootstrap, Playwright, Jenkins, GitLab

EDUCATION

B.Tech. ECE — National Institute of Science & Technology, Berhampur

CGPA 8.02 · 2015–2019

INTERESTS

Reading Team Activity Planning Event Anchoring Mentoring Long Walks